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Dear Editor,

Submission of Research Article: "Modeling electrical losses in CsPbI3 solar cells: stacked idealizations and a routine-quality envelope at 18.0%"
by Hussain Touhid Siddiquee and Md. Abdul Malek Fahim

Please consider our manuscript for publication as a Research Article in *AIP Advances*.

What the paper does. On one fully worked ITO/TiO2/CsPbI3/CBTS/Ni stack in SCAPS-1D, we (i) locate a permissive-box ceiling of 24.95% by joint optimisation of thickness, doping, defects, gap, and *both* transport-layer affinities; (ii) sample a frozen routine-quality prior volume (96-point Latin hypercube with published, measured-anchored ranges) giving a prior-predictive envelope of median 18.0% with 5–95% range 15.6–20.0%, above which the 2024–2026 experimental records fall; and (iii) price every load-bearing idealisation individually: radiative recombination, mobility, series resistance to 5 Ωcm^2, contacts, affinities, optics, and density of states (the ceiling-vs-record comparison is reported both ways because it flips with the density-of-states treatment). A pilot guardrailed-reporting checklist is proposed. The defended number is the 20.82% routine-defect result, not the ceiling.

Novelty and importance. First, joint affinity mapping replaces one-at-a-time optimisation and moves the honest ceiling from 23.50% (surveyed affinities) to 24.95%. Second, the envelope is only as good as its prior, so the prior ranges and their measurement anchors are published in the main text, not the supplement. Third, scripted resistance sweeps to 5 Ωcm^2 bound the six-point fill-factor gap against measured devices as roughly half resistive, half recombination-driven. This is the electrical-loss half of the realistic-simulation protocol of Saidarsan et al.; the wavelength-dependent (optical) half was implemented by Guruprasad et al. (*Solar Energy* 305 (2026) 114251), cited as Ref. [11].

Fit to the journal. *AIP Advances* publishes primary research across the physical sciences with rigorous technical standards and conclusions supported by presented data. This manuscript is a reproducibility study that disciplines photovoltaic performance modelling: every number traces to archived simulation receipts in the public repository (cited in the Data availability statement), the analysis meets the journal's criteria (new simulation results, full detail, clear scientific English), and the nontechnical motivation, results, and conclusions are stated for a broad audience.

Statements. This work is original, is not under consideration elsewhere, and has been approved by all authors. The authors have no conflicts to disclose. All SCAPS results come from archived simulation receipts in the public repository, so every number is reproducible. AI assistance was used for editing and consistency checking only, as declared in the manuscript; no results were generated by AI. Figures use the authors' own simulation outputs exclusively.

Suggested reviewers.

1. Dr. Thomas Unold, Helmholtz-Zentrum Berlin (perovskite doping density and device physics).
2. Dr. Joseph Luther, National Renewable Energy Laboratory (CsPbI3 quantum-dot and thin-film photovoltaics).
3. Dr. Faiz Arith, Universiti Teknikal Malaysia Melaka (SCAPS-1D perovskite device modelling).

Contact details will be entered in the submission system; the editors are of course free to choose other reviewers.

Thank you for your consideration. We look forward to your response.

Sincerely,
Hussain Touhid Siddiquee (corresponding author)